

3G and WiMAX

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India has seen rapid technological growth in recent times, especially in the areas of information technology and communication. Telecommunication has led the charge in the area of consumer technology with rapid penetration of mobile telephony amongst the masses. India is already one of the largest consumers of mobile phones, with an average nine million users being added every month. The telecom tariff rates offered in India are among the lowest in the world.

3G and WiMAX are the latest inclusions to the mobile and internet communication fold, promising to take user-experience to a higher level through interactivity, greater convenience, improved access, and quality of services offered. This article aims to capture the essence of these two technologies, analyze their impact on communication, and to conduct a comparative assessment of relevance, and hence acceptability, of the two to the masses.

3G

Launch of 3G in India

Doors are opening for the launch of the much awaited 3G service in India. The introduction

Table 1				
Frequency band	Availability	Freq block size	Uplink frequency	Downlink frequency
2.1 GHz 3	Available	2 x 5 MHz	1920-1980 MHz	2110-2170 MHz
450 MHz 4	Proposed	2 x 1.25 MHz	Undecided	Undecided
800 MHz	Proposed	2 x 1.25 MHz	Undecided	Undecided
1900 MHz	Proposed	2 x 1.25 MHz	1900-1910 MHz	1980-1990 MHz

High speed mobile connectivity is coming to India

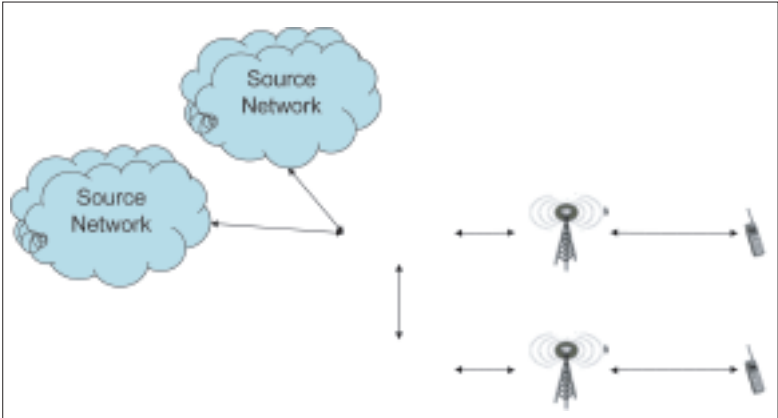
of advanced telecommunication services was proposed in the tenth five year plan, paving the way for funding and resource allocation for 3G services in the annual budget of the Department of Telecommunication (DoT). The electronic auctioning of the 3G spectrum is planned to take place in two phases on 16th January and 19th February, according to a DoT directive. This fresh initiative should pave the way for a wide range of products and mobile services being made available to the Indian consumer. 3G will facilitate high-speed internet access through mobile phones and provide mobility to a large number of services such as video conferencing, gaming, and large downloads.

Currently, the 2.1 GHz band is to be auctioned for launching 3G services in India. License winners are expected to roll out 3G services in 90 per cent of metro areas and 50 per cent in other areas given to them (with a minimum 15 per cent rural coverage). This coverage is expected by the end of the fifth year, after spectrum allocation.

The 450 MHz band is currently being utilized for radio communication, while the 800 and 1,900 MHz bands are used for military communication purposes. They will be auctioned when they become free.

What is 3G?

“Third Generation wireless telecommunication technology”, or 3G, is an advanced radio communication technology which provides a marriage of high-speed internet access and high-quality mobile telephony. The technology enables simultaneous voice and non-voice data transfer.



The Generation Gap: 3G versus its siblings				
Generation	Switching technology	Data type	Period of launch	Standards
1G	Circuit switching	Analog	1980	AMPS
2G	Circuit switching	Digital	Early 90s	"TDMA, CDMA, GSM"
2.5G	Circuit switched for access network & packet switched for core network	High speed digital	Early 21st century	"GPRS, EDGE, 1xRTT"
3G	Packet switched	Digital broadband	2003	"WCDMA, CDMA-2000"

1-G technology involves the use of analogue radio signals for communication. Voice data is modulated to a high frequency (usually beyond 150 MHz) for transmission. Information transfer between telephone towers takes place in the form of digital signals.

2-G differs from 1-G in the use of digital voice signals for communication. The analogue voice data is encoded into digital signals for transmission.

Differences between the three technologies		
1G 1G technology was the starting block in mobile telephony offering quality voice service	2G 2G technology brought in energy efficient digital communication systems facilitating smaller handsets and price reduction	3G "Launched in 2001, the 3G services have brought in high quality service and substantial user convenience, but involve considerable investment in technology for service providers"
Analogue voice service	Digital encryption of conversation	High speed internet on phones
Better sound quality than 2G over long distance calls	Better data security	Video communication
No data service	Optimal spectrum utilization	Heavy data transfer services
	Start of data services like SMS	Quality and interactivity of services
	Limited e-mail access	Enhanced roaming
	Slow speed internet access	Strong network security
	14.4-43.2 kilobits/sec data transfer	144 Kbits/sec (high speed mobile) & 2 Mbits/sec (stationary mobile) 7

How does 3G compare with the available technology?

3G is a substantial enhancement over the current 2G technology, and is capable of facilitating high-speed internet access and data transfer. Adoption of 3G in India would involve substantial improvement in the existing cell phones to be able to support 3G services. The idea of launching 2.5G services was also considered due to the difficulty of moving over from 2G to 3G. But the delay in launching advanced telecom services and launch of phones with 3G capabilities made the choice easier.

High speed digital packet switching networks are the source of the high speed offered by 3G. Compared to the traditional circuit switching technology, packet switching offers better transmission bandwidth utilisation, and hence higher data transfer rates. Additionally, advance multiplexing techniques add to network speed.

Packet switching vs. circuit switching

Circuit switching involves dedicated link creation between two phones during a call leading to redundancies & high waiting time

In **packet switching** data is transmitted in the form of encrypted digital data packets through the least congested route / nodes reducing redundancy in the process

What changes can be expected due to 3G?

The emergence of 3G is likely to act as catalyst for enhanced product and services. Creative solutions evolving around user benefits are expected to emerge.

- Quality of text, audio and video communication would improve through:
 - Technology supporting more real-time, and next to real quality communication
 - Enhanced mobility and roaming
 - High quality e-mail and internet access
- 3G would enable heavy video broadcast (video conferencing, video lectures, etc.) and virtual classrooms, video sharing, etc.) and data-intensive services (such as online monitoring/surveillance, e-commerce, online education, telemedicine, music/software/data download, online gaming, etc.)
- High-end, light, feature intensive, and easy-to-use terminals would come into the market with creative interfaces designed to suit customer preferences.

Additionally, the strong business opportunities around 3G are expected to attract a large number of players offering choice, affordability and quality to the consumer.